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11 June 1984
SUPERSEDING
MIL-T-21900E(YD)
10 December 1979

MILITARY SPECIFICATION

TRUCK, TANK; AIRCRAFT LUBRICATING OIL DISPENSING, 500 GALLON, 4 BY 2

This specification is approved for use by the Naval Facilities Engineering Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a commercial, gasoline-engine-driven, four wheel, two rear wheel drive, 500-gallon capacity, aircraft lubricating oil dispensing truck.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

W-B-131 - Battery, Storage, Vehicular, Ignition Lighting, and Starting.
O-E-915C - Extinguisher, Fire, Dry-Chemical (Portable).

MILITARY

MIL-N-7284C - Nozzle, Oil Servicing, Pistol Grip, Nondrip, Type A-24.
MIL-L-22851C(1) - Lubricating Oil, Aircraft Piston Engine, (Ashless Dispersant).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 2320

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STANDARDS

FEDERAL

FED-STD-595 - Colors.

MILITARY

MIL-STD-281 - Automobiles, Trucks, Truck-Tractors, and Trailer Dollies; Preservation and Packaging of.
MIL-STD-1223 - Non Tactical Wheeled Vehicles Treatment, Painting Rustproofing, Identification Marking, And Data Plate Standards.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents form a part of this specification to the extent specified herein.

CODE OF FEDERAL REGULATIONS (CFR)

Title 49 - Transportation, Part 178.

DEPARTMENT OF TRANSPORTATION (DoT)

Federal Motor Vehicle Safety Standards and Regulations.
Federal Motor Carrier Safety Regulations.

NATIONAL BUREAU OF STANDARDS (NBS)

H44 - Specifications, Tolerances, and Regulations for Commercial Weighing and Measuring Devices.

(Application for copies of CFR, DoT, and NBS publications should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

ENVIRONMENTAL PROTECTION AGENCY (EPA)

Code of Federal Regulations, Title 45, Subtitle A, Part 1201 - Motor Vehicle Air Pollution Standards.
Interstate Motor Carrier Noise Emission Standards.

(Application for copies should be addressed to the Public Affairs Office, Environmental Protection Agency, Rockville, MD 20852.)

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2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM B209 - Sheet and Plate Aluminum and Aluminum Alloy.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

SAE Handbook

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

STATE OF CALIFORNIA

California Vehicle Code.

(Application for copies should be addressed to the Department of Motor Vehicles, 2570 24th Street, Sacramento, CA 95809.)

TIRE AND RIM ASSOCIATION, INC. (TRA)

TRA Yearbook.

(Application for copies should be addressed to the Tire and Rim Association, Inc., 3200 West Market Street, Akron, OH 44313.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The truck shall be a gasoline-engine-driven, four wheel, two rear wheel drive chassis with cab, of not less than 16,000 pounds (lb), gross vehicle weight (gvw), equipped with a single compartment aluminum alloy tank of 500-gallon capacity. The truck shall be equipped with pumping system, meter, oil heating system, and all components required to make a complete aircraft lubricating oil dispensing system.

3.2 First production vehicle. The contractor shall furnish a truck for first production vehicle inspection.

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3.3 Standard truck, components, and accessories. Except as specified in 3.3.1 through 3.3.7, the truck, components, and accessories shall be standard or optional items which meet or exceed the requirements of this specification. All items shall be as represented and rated in the manufacturer's sales information, including special or mounted equipment. Sales information shall be limited to specifications and technical material, identical to that furnished to the authorized company representative for selection of truck models and components. Such items shall be construed as commercial motor vehicle products, and shall comply with all DoT Federal Motor Vehicle Safety Standards and Regulations applicable, on the date of manufacture.

3.3.1 Heavy-duty cooling system. A heavy-duty cooling system shall be furnished that shall maintain engine coolant at a temperature below the boiling point with truck loaded to rated gvw, and operated at an altitude of 10,000 feet above sea level or in an ambient air temperature of not less than 125 degree Fahrenheit (°F).

3.3.2 Coolant system and indicators. Coolant system shall include a surge tank, a deaeration system, or shall include a coolant recovery reservoir of not less than 2 quart capacity. In addition, a low coolant level or high coolant temperature alarm buzzer and red indicator warning light shall be provided on the cab instrument panel.

3.3.3 Towing devices. Not less than two hooks or loops for towing the truck shall be furnished on the front of the truck. Mounting shall include reinforcement to transfer loads directly to chassis rails.

3.3.4 Radio noise suppressors. The truck chassis shall be equipped with radio noise suppressors conforming to SAE J552. Electromagnetic radiation from the truck chassis shall be within the limits of SAE J551.

3.3.5 Sound level. The exterior sound level produced by the truck shall not exceed the noise limits established by the EPA, in EPA Noise Emission Standards for Transportation Equipment, Medium and Heavy Trucks. The interior sound level shall be in accordance with DoT Motor Carrier Safety Regulation, section 393.94.

3.3.6 Splash guards and shields. Rear wheels shall have metal splash shields ahead of the wheels and splash guards to the rear. Splash and stone throw protection shall be in accordance with SAE J682.

3.3.7 Engine hour meter. An engine hour meter, having a totalizing mechanism of not less than 9,999 hours, shall be furnished for the truck chassis engine to register accurately the number of hours of operating time. The meter shall be of rugged construction to insure continuous trouble-free performance under severe operating conditions. Engine hour meter shall be mounted on the cab instrument panel.

3.4 Air pollution control. Trucks shall comply with the EPA regulations governing Control of Air Pollution from New Motor Vehicles and New Motor Vehicle Engines in effect on the date of manufacture. In addition, trucks destined for California shall comply with State of California regulations governing air pollution control in effect on the date of manufacture.

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3.5 Weights, ratings, and dimensions.

3.5.1 Curb weight. Curb weight shall include weight of chassis and cab, with all attachments, accessories, equipment, body, full complement of fuel, lubricants and coolant.

3.5.2 Weight rating. The gvw rating shall consist of curb weight, operator (weight computed at 175 lb), and payload to provide not less than 16,000 lb gvw.

3.5.3 Weight distribution. The distribution of the gvw for the purpose of establishing suspension, axle, and tire capacities shall be determined with the payload uniformly distributed over the load area.

3.5.4 Ratings. Component and vehicular ratings shall not be raised to meet the requirements of this specification. Minimum gvw shall be 16,000 lb.

3.5.5 Cab-to-axle dimension. The chassis shall have a cab-to-axle dimension of not less than 60 inches.

3.6 Performance.

3.6.1 Speeds. High and low speed requirements shall be met with the truck loaded to specified gvw.

* 3.6.1.1 High speed gradeability. The truck shall ascend a continuous grade of 2.5 percent at 50 miles per hour (mph). Gradeability requirements shall be met with the main transmission in direct drive. Gradeability shall be verified with calculations in accordance with SAE J688 (see 6.3).

3.6.1.2 Low speed. Low speed shall be calculated with engine operating at not less than 35 percent of recommended governed speed, and shall not exceed 5 mph.

* 3.6.1.3 Maximum geared speed. Maximum geared speed at engine governed speed shall be not less than 55 mph. Conformance to geared speed specified shall be determined by calculating in accordance with the following formula:

$$\text{Maximum geared speed (mph)} = \frac{\text{Governed speed (rpm)}}{\text{Total gear reduction X Tire factor}}$$

* 3.6.2 Service brakes. Service brakes shall control and hold truck, when loaded to its specified gross weight, on a 30 percent grade. The service brakes shall stop the vehicle, loaded to specified gvw, within the stopping distance requirements of Federal Motor Carrier Safety Regulation 393.52.

3.7 Chassis components.

* 3.7.1 Engine. The truck engine shall be a liquid cooled, four-stroke cycle, gasoline engine, having horsepower (hp), torque, and speed characteristics to satisfactorily meet all truck performance requirements

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specified herein. The engine furnished shall produce the required truck performance when operated on unleaded gasoline having a research octane rating of 91. The engine shall be capable of continuous, warranted operation on unleaded fuel. Maximum net engine torque shall be not less than 250 lb feet. Engine net torque rating and net hp used in performance production calculations shall be determined in accordance with SAE J245.

3.7.1.1 Oil filter. Full flow type oil filter shall be furnished.

3.7.1.2 Governor. Engine governor shall be furnished and set and sealed to limit engine to engine manufacturer's maximum recommended operating speed.

3.7.1.3 Powerplant heaters. When specified (see 6.2), coolant, engine oil, and battery heaters shall be provided. Heaters shall operate on 110-volt (V) alternating current (ac), and shall be wired through a junction block to a single three pronged (male), weatherproof, slave receptacle for receiving external power and grounding the truck. A three wire connecting cable, 25 feet long and of adequate line capacity to supply power for all heater units simultaneously, shall be furnished. Connecting cable shall include a matching female connector at the truck end and a standard, weatherproof, three pronged (two power plus one ground) male connector at the other end. Electrical apparatus shall conform to DoT Motor Carrier Safety Regulations, section 393.77. Electrical insulation of connecting cable shall withstand normal operating stresses in low ambient air temperatures (down to -60°F) without cracking or loss of dielectric capacity. All heater lead wires shall be installed without interfering with truck component operation, and without loose excess wire. A carrier for the connecting cable shall be mounted within the cab or engine compartment and shall provide positive cable retention during truck operation. Heaters shall be furnished as follows:

a. Coolant heater, 1500-watt minimum rating, shall be installed in the engine block or lower, coolant, inlet hose. A coolant circulating pump, driven by a 110V alternating current motor, shall be provided. Engine thermostat, minimum 195°F, shall be installed.
b. Immersion type engine oil heater, 300-watt minimum rating, with minimum 195°F thermostat, shall be installed in oil pan through any convenient opening.
c. Battery heater shall have a capacity adequate to maintain the battery electrolyte at a temperature of not less than 10°F during truck exposure in ambient air temperatures as low as -60°F, and shall embody a thermostat to limit the temperature of the electrolyte to not more than 80°F.

3.7.2 Electrical equipment.

* 3.7.2.1 Ignition system. The truck shall be equipped with a 12V ignition system. An alternator of not less than 40 amperes rated capacity, and which provides not less than 16 amperes direct current output at normal engine idle speed, shall be furnished.

3.7.2.2 Lighting and wiring. The truck lighting and wiring shall conform to and be installed in accordance with DoT Motor Carrier Safety Regulations,

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sections 393.12, 393.19, 393.20, 393.22 through 393.31, 393.33, and as specified herein. All wiring in the vicinity of the tank, pumping compartment, and rear cabinet shall be enclosed in copperized or aluminum tubing, which shall be connected to vaporproof junction boxes with compression fittings.

3.7.2.3 Turn signals. Turn signal units conforming to SAE J588 with SAE J589, class A operating units mounted on the steering column, shall be furnished and shall be provided with double faced front signal units and with single faced rear signal units installed in accordance with SAE J588, except that turn signals shall not be mounted on engine compartment hood. Turn signal operating units shall be self-canceling type with visible and audible flash indicator.

* 3.7.2.4 Battery. The truck shall be equipped with a 12V battery. The total reserve capacity rating shall be not less than 110 minutes. The total cold cranking rate at 0°F shall be not less than 450 amperes. Reserve capacity ratings and cold cranking ratings shall be measured in accordance with SAE J537. Batteries shall be of the maintenance-free type, having the maintenance-free characteristics listed in W-B-131. The battery shall not be located in the cab.

3.7.3 Fuel system. Fuel system shall conform to DoT Motor Carrier Safety Regulations, section 393.65 and 393.67.

3.7.3.1 Air cleaner. An air cleaner shall be furnished. When equipped with an oil bath type, oil capacity shall be not less than 1 quart.

3.7.3.2 Fuel tank. Fuel tank shall be manufacturer's standard for the truck furnished with not less than 30 gallons total capacity. When more than one tank is furnished, a selector valve connecting either tank to engine fuel intake shall be provided.

* 3.7.4 Exhaust system. The exhaust system shall conform to DoT Motor Carrier Safety Regulation, section 393.83. A spark arrestor shall be provided. All horizontal portions of the exhaust system, which project to the rear of the cab, shall be provided with a cover plate. Cover plate shall form a shield that will shed fluids, precluding spilled fuel from coming in contact with the exhaust system. Exhaust system, when stacked, shall be furnished with a heat shield and a hinged rain cap.

* 3.7.5 Transmission. A manually shifted transmission shall be provided. Input torque capacity of transmission shall be at least equal to the maximum torque delivered by the engine. The transmission shall have not less than four forward speeds and one reverse speed. Transmission shall provide for maximum ease of shifting with synchronized shift for at least the two highest forward speeds. The transmission shall be provided with SAE J704 power take-off openings at right and left sides. The transmission and axle gear ratio shall be selected to provide the performance specified in 3.6.1.

* 3.7.5.1 Power take-off. A power take-off shall be furnished to operate powered equipment. Controls to operate the power take-off shall be located in

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the truck cab accessible to the seated driver. A caution plate or decal reading "DO NOT OPERATE VEHICLE AT HIGHWAY SPEEDS WITH POWER TAKE-OFF ENGAGED" shall be provided and installed so as to be readily visible to the seated driver.

3.7.6 Clutch. Clutch shall be the largest capacity clutch offered for the truck furnished with torque exceeding maximum delivered engine torque.

3.7.7 Drive line components. Drive line components shall be adequate to transmit the maximum delivered torque of the engine, as developed through the maximum gear train reduction.

3.7.8 Frame. Chassis frame shall be manufacturer's standard for the truck furnished.

3.7.9 Suspension. Truck shall be equipped with suspension system with components having a rated capacity at least equal to the load imposed on each member, measured at the ground, with truck loaded to rated gvw. When suspension capacity is rated at the spring pads, unsprung weight shall be deducted. Truck shall be equipped with hydraulic, double-acting shock absorbers at the front wheels.

* 3.7.10 Axles. Axle ratings shall be at least equal to the load imposed on each axle, measured at the ground, with truck loaded to rated gvw. A single reduction rear axle shall be provided.

3.7.11 Traction control. The vehicle shall be furnished with traction control on the rear axle. The traction control shall actuate automatically to insure that power is transmitted to the wheel having traction when the opposite wheel loses traction. Maximum traction capabilities shall be maintained at all times under each drive wheel for the life of the vehicle.

3.7.12 Brakes. Brakes shall conform to DoT Motor Carrier Safety Regulations, section 393.40 through 393.43, 393.45 through 393.52, and as specified herein.

3.7.12.1 Service brakes. Truck shall be equipped with power assisted, hydraulic actuated, four wheel service brakes.

3.7.12.2 Parking brake. Truck shall be equipped with a parking brake.

3.7.13 Emergency shutdown switch. An emergency shutdown switch, capable of stopping the truck engine, shall be mounted on the pumping assembly control panel.

3.7.14 Power takeoff lockout. An interlock system shall be provided to prevent simultaneous propelling of the truck and operations of the pumping equipment. The interlock system shall be mechanical or electrical type which shall lock up the service brakes when power takeoff is engaged.

3.7.15 Wheels, rims, tires, and tubes. Truck shall be equipped with single front and dual rear wheels. Rim and tire ratings shall conform to TRA recommendations, for the type and size of tires furnished. Tire and rim sizes shall be the same for all wheels on each truck. Disk type wheels shall be provided.

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3.7.15.1 Tires. Tires shall be tube or tubeless type with highway tread. Tires shall be of rated capacity at least equal to the load imposed on each tire, measured at each wheel, at the ground, with truck loaded to rated gvw.

3.7.15.2 Tubes. Tubes, when furnished, shall be of heavy-duty type, and shall be of proper size for tires furnished.

3.7.15.3 Tire chain clearance. Clearance shall be provided for installation and use of tire chains.

3.7.15.4 Spare wheel. A spare wheel or rim shall be mounted on a carrier and installed in a readily accessible location on the truck.

3.7.16 Cab. Manufacturer shall furnish a standard full-width cab. Cab doors shall be equipped with locks, operable from inside the cab through mechanical linkage, with at least the curbside door equipped with external, key-operated lock. Cab shall have upholstered, full-width, adjustable seat and back or individual, adjustable, driver's seat and individual passenger seat. Interior dome lighting shall be provided. Manufacturer's standard fastenings and two pairs of seat belts shall be furnished and installed. Drip rails shall be installed above the cab doors. The color of the upholstery and the interior finish shall be compatible with the exterior color (see 3.11). White upholstery shall not be furnished.

3.7.17 Windshield wipers and washers. Truck shall be equipped with dual windshield wipers and windshields washers. Windshield wipers shall be multi-speed type and operated by electric motor(s).

3.7.18 Steering. Manufacturer's standard power steering shall be furnished.

3.7.19 Heater and defroster. Hot water heater shall be provided. Heater shall have fresh air intakes. Discharge outlets shall be provided to direct heated air to floor and to defroster louvers. Heater shall be complete with blower and mounted controls convenient to the driver.

3.7.20 Front bumper. A channel type front bumper shall be provided.

3.7.21 Rear end protection. Rear end of truck shall be protected in accordance with DoT Motor Carrier Safety Regulations, section 393.86. A rear bumper shall be provided.

3.7.22 Controls and operating mechanism. All controls and operating mechanisms shall be located for left-hand drive. Controls shall be complete and conveniently operable by the driver. Lever controls shall be designed and located to permit easy entrance and exit of operator to and from driver's compartment. Instruments and controls shall be identified as to their function and installed in a manner to facilitate removal and servicing. Instruments shall be panel mounted.

3.7.23 Accessories and equipment. Chassis equipment shall be complete with all accessories furnished as standard equipment by the manufacturer. The following minimum equipment shall be furnished:

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a. Key-operated ignition switch.
b. Ammeter, charging indicator or voltmeter.
c. Fuel gage.
d. Oil pressure gage or indicator.
e. Engine temperature gage or indicator.
f. Speedometer with recording odometer.
g. Dual sunvisors.
h. Front door mounted armrest on driver's side.
i. Driver's compartment ventilator other than window.
j. Tachometer (see 3.8.3.3).
k. Manufacturer's standard electric horn.

3.7.24 Outside rearview mirrors. Combination (flat and convex) outside rearview mirrors, enclosed in a common housing, shall be provided on each side of cab. Each combination mirror shall have at least 50 square inches of flat reflective area and an adjustable convex surface having at least 20 square inches of reflective area. Mirrors shall have not less than three supporting arms.

3.8 Body components.

3.8.1 Tank. The tank shall be of single compartment type, with not less than 500-gallon capacity, plus at least a 3 percent expansion space. The tank shall be constructed of 5,086 series alloy sheet aluminum, conforming to ASTM B209, of sufficient thickness to provide complete structural integrity of the tank and meet performance and other requirements specified herein. Design and construction of the tank shall conform in all respects to the requirements of CFR 49, paragraph 178.341, specification MC306. The tank shall be shaped with oval lower section and flat on top. The tank flat top shall be covered with diamond plate or other antiskid material. The flat top shall be reinforced, if necessary, as it will be used as a platform for oil servicing of air craft. The height of platform shall be not over 78 inches above ground level with tank empty.

3.8.1.1 Manhole. The tank shall have a circular manhole, with cover, with a diameter of not less than 16 inches. The manhole cover shall have a 10 inch, watertight, hinged, filler cover with necessary safety features to eliminate the possibility of a spark occurring during opening or closing of the cover. The manhole shall be equipped with a buna-N gasket, or equivalent, to insure a fuel tight seal between the collar and the cover. Cover shall be equipped with spring loaded, quick-release latches or fusible plugs to provide additional emergency pressure release. The fill cover shall be so installed that the hinge faces the cab so that if the lid is left open, it immediately falls into position and locks automatically when truck starts.

3.8.1.2 Tank top guard. A tank top guard, at least 2-1/2 inches high, to provide protection against personnel slipping off tank top shall be mounted on all edges of the tank top platform. The design of this guard shall be such that it will not trap water or spilled oil.

3.8.1.3 Overturn protection. Tank top equipment, such as fill cover,

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manhole, and piping connections, shall be provided with suitable protection in event the tank overturns. Overturn protection shall not have closed ends or other construction features that will trap spilled product or water accumulation.

3.8.2 Rear cabinet. Rear cabinet shall be provided and constructed so as to adequately support the equipment therein without undue vibration. In addition, the cabinet shall be so constructed as to be weather- and splash-proof and shall be provided with drains. The cabinet doors shall be suitably stiffened and adequately supported with side hung, flush, continuous type hinges containing 3/16-inch minimum diameter brass or stainless steel rods. Flush type hardware, including positive-type latches, cadmium plated or otherwise processed to prevent corrosion, shall be provided. Latches to hold doors in open position shall be provided. A container, suitable for dry storage and protection of instruction manuals, shall be attached to some accessible area inside the rear cabinet or truck cab or glove compartment. Material shall be steel and prominently labeled: "INSTRUCTION MANUALS". The top of the cabinet shall be flat and shall be part of the top platform, and shall be suitably reinforced. A means shall be provided to allow the hose to extend through the cabinet top, without removal of the nozzle. A bimetal temperature gage with not less than 2-1/2 inch dial, range -40°F to +120°, shall be mounted in rear cabinet with the temperature sensing element extending into the rear of the tank.

3.8.3 Pump, pump drive, and control.

3.8.3.1 Power takeoff. One heavy-duty power takeoff shall be furnished with the lowest speed ratio available, capable of transmitting sufficient hp for satisfactory pump operation.

3.8.3.2 Power takeoff control. In order to prevent operation of the truck when the power takeoff and the pump are connected, the power takeoff control shall be interconnected with the service brakes. An electric or mechanical device shall be incorporated into the power takeoff control which will lockup the service brakes when power takeoff is engaged. The service brake lockup shall not release until power takeoff is disengaged and in the neutral position. A red indicator light shall be provided and shall be illuminated when power takeoff is engaged and shall be properly labeled as to its function. A caution plate reading, "Service brakes are applied when engaging power takeoff. Do not attempt to move vehicle with power takeoff engaged." The power takeoff control shall be so located that it may be operated by the driver when sitting in the driver's seat.

3.8.3.3 Engine tachometer. An electric tachometer (see 3.7.23(j)), with optimum engine speed red-lined for satisfactory pumping operation, shall be provided and shall be so located on the truck cab dash as to be visible when operating the hand throttle control.

3.8.3.4 Pump. A rotary or gear type positive displacement pump capable of withstanding a working pressure of 200 pound-force per square inch (psi), and a test pressure of 300 psi shall be provided. The pump shall produce a flow rate at the nozzle of not less than 10 gallons per minute (gpm) when pumping

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aircraft lubricating oil conforming to type IIA of MIL-L-22851C(1) at a temperature of not more than 65°F. The flow rate at the nozzle shall be not less than 16 gpm when pumping this oil at a temperature of not more than 85°F. If required, an oil enclosed gear reduction assembly shall be provided to achieve performance specified. The pump propeller shaft shall have two universals and slipjoints if the shaft is 48 inches or less in length, and shall have a self-aligning bearing center bracket and three universals if over 48 inches in length. The universals shall have as little angularity as possible. Propeller shaft and universals shall be of ample size to carry the load. The pump shall be *mounted* on a cradle supported from both chassis side members with supporting bolts close to side axis. The pump shall be located as close to tank outlet as possible and as high as practicable to provide ample ground clearance. The pump cradle shall not be welded to frame side members. Pump shall be capable of pumping aircraft lubricating oil from ground level to the 500 gallon dispensing tank on the truck.

3.8.3.5 Pressure gage. A pressure gage with shutoff cock, scale 0 to 300 pounds per square inch gage, shall be provided and connected to pump outlet piping and shall be located in rear cabinet. The gage shall incorporate a means to dampen pressure oscillations to afford stability of the indicating needle.

3.8.3.6 Emergency valve. A flanged-type, 4-inch, low level seat, emergency shutoff valve, constructed of aluminum, with 1/4-inch mesh monel wire screen, shall be provided with open mesh area 50 percent larger than the area of the 4-inch emergency valve. The emergency valve shall be located in the tank on the suction line to the pump. The emergency valve remote control shall be located within reach of the driver, seated in the driver's seat and shall be labeled: "EMERGENCY VALVE MUST BE OPENED BEFORE ENGAGING POWER TAKEOFF". Fusible links shall be incorporated in the emergency valve or its operating control to provide automatic closing of the emergency valve in the event of fire. A second aluminum emergency valve, not less than 3 inches in nominal diameter, shall be provided in the tank where the return line from the pressure relief valves enter the tank. This emergency valve shall be of the check valve type, shutting off all flow from the tank into the return line. The valve shall be spring loaded, set to open and permit oil to enter the tank through the return line whenever the 45 psi pressure relief bypass valve opens.

3.8.4 Piping and control valve system. All piping shall be stainless steel tubing with wall thickness to withstand 200 psi working pressure and 300 psi test pressure. Piping and valves shall be sized to provide the performance flow rates specified. Valves shall be cast iron, or brass, or aluminum. A pressure relief bypass valve adequate for 45 psi shall be provided.

3.8.4.1 Suction stub piping system. A suction stub with dust cap, control valve, and 60 mesh strainer shall be provided for pumping from an outside source and shall be located on the driver's side (left front of tank) of the vehicle. The suction stub shall also be used for draining of the tank by gravity. The strainer shall be located upstream from the suction stub control valve.

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3.8.4.2 Piping connections. Groove-type couplings with two half clamps and synthetic gaskets shall be provided where necessary to relieve piping strain, and all pipes and manifolds shall have companion flange couplings with suitable gaskets or groove-type couplings, where necessary, to facilitate disassembly for repair and maintenance. Piping arrangement shall allow for disassembly of all piping without removal of other components.

3.8.5 Drum suction hose and pipe. A drum unloading suction hose and pipe shall be provided with provisions for stowing in rear cabinet with holddown clips. The hose shall be suction type, 8 feet long, 2-inch size. One end shall have a 2-inch quick coupler with dust plug and security chain and the other end shall be 2-inch brass male coupling. The suction pipe shall be brass, 1-1/2 inch size, 40 inches long, notched at lower end with pipe thread at top, coupled to a 1-1/2 by 2-inch brass reducing elbow. The unit shall be assembled.

3.8.6 Heating system. When specified (see 6.2), a method of heating the aviation lubricating oil shall be provided. The tank shall be provided with 110V, 60 cycle, single phase strip heaters, which shall maintain oil temperature at not less than 40°F when operating in an ambient temperature of 0°F. Wiring shall be in conduit. Thermostat shall be furnished and located in the rear compartment. Strip heaters shall be so wired that in the event any circuit should fail, the other circuits shall be operating. Each circuit shall be provided with a means to indicate that each circuit is operating. A red indicator light shall be provided to indicate when heater switch is on.

3.8.7 Meter. A 2-inch positive displacement meter, with a maximum rating of not less than 24 gpm based on aircraft lubricating oil conforming to MIL-L-22851C(1), type IIA, without strainer or air eliminator, shall be housed in rear cabinet. The meter shall be installed downstream of the 45 psi pressure relief bypass valve. Meter shall be designed for 75 psi working pressure and 125 psi hydrostatic test pressure. Meter shall have an accuracy in accordance with NBS H44. Meter shall be capable of being adjusted, while under pressure, without leakage or loss of product. A vertical meter counter shall have not less than a seven figure totalizer indicating full gallons and a four or five figure setback indicating full gallons. Figures in the setback shall be not less than 3/4-inch in height, painted with light-reflecting paint. A clock-type counter will not be acceptable. The meter counter shall be mounted on a rigid extension through the top of the rear cabinet, allowing the counter to be rotated in order that the face may be visible from any direction.

3.8.8 Filter. A lubricating oil filter having a selectivity equal to at least a 60 mesh screen shall be provided. No paper or material other than cellulose type material shall be used in the filter or filter elements. The element shall be capable of withstanding a 300 psi differential pressure without rupture or mechanical failure. The filter shall be installed upstream of the meter, upstream of the 45 psi pressure relief bypass valve, and in a manner that will facilitate replacement of the filter elements without excessive spillage of product in the equipment box.

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3.8.9 Hose reel. A hose reel of nominal 2-inch size shall be provided in the rear cabinet and shall have a capacity of 50 feet of 1-1/2 inch hose. The 2-inch inlet and the 1-1/2 inch outlet on the hose reel shall accommodate the 1-1/2 inch hose.

3.8.9.1 Hose. Thirty feet of 1-1/2 inch hose shall be furnished. Hose shall accommodate aviation lubricating oil specified herein.

3.8.9.2 Nozzle. The hose shall terminate in a 1-1/2 inch nominal size nozzle, which shall conform to the requirements of MIL-N-7284C, modified with the following supplementary provisions:

a. Nozzle spout shall be 1-1/2 inch internal diameter.
b. Nozzle spout shall provide a means of venting between the nozzle spout and the aircraft filler neck of 1-7/8 inch opening.
c. Length of nozzle spout shall be increased 6 inches.

3.8.10 Body mounting. Body shall be secured with U-bolts, twin studs, or by brackets, and shall include a breaker strip.

3.8.10.1 U-bolts or twin studs. When U-bolts or twin studs are used, there shall be not less than four U-bolts or twin studs, per side, each having 9/16-inch minimum body diameter with 5/8-inch minimum thread diameter. Tie-plates shall be at least 1/2-inch thick. Vehicle chassis frame shall be braced, using wood blocks at each mounting point unless mounting point is located at a full depth, frame crossmember. Blocks shall incorporate a keeper strap or groove for mounting bolt, and shall be of a width and thickness to assure retention. Two tie back straps shall be provided, one on each side of rear portion of body sub-frame, to maintain body alinement on vehicle chassis. Forward body mounting bolts shall be located to the rear of the tapered portion of breaker strips (see 3.8.10.3).

3.8.10.2 Brackets. When brackets are used, they shall be bolted to the web of the chassis frame rails. The body mounting brackets shall provide means of drawing down the body on the chassis rails and provisions shall be made to prevent lateral shifting of the breaker strips. When additional holes are required to secure mounting brackets to chassis frame rails, they must be located within the area of the rail which is designated as being safe for drilling in accordance with the chassis manufacturer's body builders layouts. Attachments shall not interfere with or obstruct chassis components.

3.8.10.3 Breaker strips. A hardwood or dense southern yellow pine breaker strip, of not less than 1-1/16 inches finished thickness, shall be installed between longitudinal sills and vehicle chassis frame. Breaker strips shall have a taper of 1 inch in 18 inches at the forward end.

3.8.11 Steps and ladders. A means for stepping from the ground to the right front of the tank platform shall be provided. A means for ascending from the ground to the left rear of the tank platform shall be provided. All steps or ladders shall be provided with suitable grab handles in order that personnel may ascend to and descend from the tank platform in a safe manner. Treads of all ladders and steps and walking surfaces of all walkways shall be of an antiskid type.

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3.8.12 Cabinet light. One or more vaporproof and shatterproof lights shall be located in the rear cabinet to provide light for the cabinet interior so that instruments installed in cabinet may be clearly read. Waterproof and vaporproof wiring shall be used exclusively. The voltage of all lamps shall coincide with the truck chassis.

3.8.13 Fire extinguisher brackets. Brackets for two 30 lb capacity dry chemical fire extinguishers conforming to O-E-915C, type I, shall be installed in a location from which the extinguisher can be readily released in one operation with one hand when a person is standing on the ground.

3.8.14 Grounding. All metal parts and units shall be bonded together to prevent the possibility of a spark due to accumulation of static electricity.

3.8.15 Instruction plates. The truck shall be equipped with instruction plates suitably located, describing any special or important procedure to be followed in operating and servicing the equipment. Plates shall be of a material which will last and remain legible for the life of the equipment, and shall be securely affixed with nonferrous screws or bolts of not less than 1/8-inch diameter.

3.8.16 Piping diagram. An etched or embossed plate of corrosion-resistant material showing piping diagram and instructions for operation shall be attached to rear door of cabinet in a position to permit easy study.

3.9 Lubrication. Lubrication means shall be provided for all parts of the equipment normally requiring lubrication. Where the use of high lubrication pressure will damage grease seals or other parts, fittings with pressure release shall be used. The truck shall be conspicuously tagged to identify the lubricants and their temperature range.

3.10 Identification marking and data plates. Identification marking and data plates shall be in accordance with MIL-STD-1223 as required for the Navy.

3.11 Cleaning, treatment, and painting. Cleaning, treatment, and painting shall be in accordance with MIL-STD-1223, as required, for the Navy. Exterior color of the truck shall be yellow color No. 13538 in accordance with FED-STD-595. The interior cabinet surface shall be painted the same color as the exterior surfaces (see 3.7.16).

3.12 Rustproofing. When specified (see 6.2), the truck cab, chassis, and chassis underside shall be rustproofed in accordance with MIL-STD-1223.

* 3.13 Servicing and adjusting. Prior to acceptance of the truck by the Government inspector, contractor shall service and adjust each truck for operational use including at least the following: focusing of lights; adjustment of the engine, electrical and brake system; alinement of front wheels; inflation of all tires; and complete lubrication of body, equipment, chassis, engine, running gear, and servicing of cooling system with a solution of ethylene glycol and water in equal parts by volume.

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3.14 Workmanship.

3.14.1 Metal fabrication. Metal used in the fabrication of equipment shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the metal. Shearing and chipping shall be done neatly and accurately. All bends of a major character shall be made by controlled means to insure uniformity of size and shape.

3.14.2 Bolted connections. Boltholes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

3.14.3 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.14.4 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

4. QUALITY ASSURANCE PROVISION

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

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4.2.1 First production vehicle inspection. The first production vehicle produced under the contract shall be inspected by the contractor at his plant under the direction and in the presence of Government representatives. This inspection shall include the examination of 4.3 and the tests of 4.4. The purpose of the inspection shall be to determine vehicle conformity with the requirements of the contract. Acceptance of the first production vehicle shall not constitute a waiver by the Government of its right under the provisions of the contract.

4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.3, the tests of 4.5, and the packaging inspection of 4.6.

4.3 Examination. Each truck shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.4 First production vehicle tests.

4.4.1 Cleaning. Prior to the performance of any test, the lubricating oil dispensing system shall be carefully inspected for evidence of contaminants in the tank and pumping system, such as loose, spattered, or excess weld or solder chips and other foreign material. Oil dispensing system shall then be thoroughly flushed by circulating stoddards solution using the installed pump and piping, through the tank and piping system. The tank and piping system, including the pump, shall then be completely drained of all the stoddards solution and entire system flushed with filter elements intact. Prior to shipment, filter elements shall be replaced with unused ones.

4.4.2 Pressure tests. The tank of the lubricating oil dispensing truck shall be subjected to an internal static air pressure of 3 psi for a period of not less than 30 minutes. A soap solution shall be applied to the exterior of the tank and around the manhole and filler cover to facilitate detection of leakage. The oil pumping system, from the suction side of the oil pump to the hose nozzle, shall be subjected to a hydrostatic pressure of 75 psi for not less than 30 minutes. Evidence of leakage under any of the pressure tests designated above shall be cause for rejection.

4.4.3 Weight and fill check. The truck shall be weighed with the tank empty to determine curb weight and distribution of curb weight on front and rear axle. Then the tank shall be filled with 500 gallons of aircraft lubricating oil. There shall be a 3 percent minimum expansion space remaining in the tank. The gvw (including operator's weight) imposed on front and rear axles shall be utilized to ascertain that the suspension, axles, and tires furnished are of adequate capacity to meet specification requirements.

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4.4.4 Road tests. The truck, loaded to provide not less than rated gvw, shall be driven at speeds up to 50 mph over improved roads for not less than 30 miles, and at speeds of 20 mph over unimproved roads for not less than 20 miles. Upon completion of the test, the truck and equipment shall be examined for leaks, warping, distortion, or malfunction.

4.4.5 Service brake test. Service brakes shall be tested to verify conformance to the requirements of 3.6.2.

4.4.6 Flow rate. Not less than 500 gallons of aircraft lubricating oil, conforming to MIL-L-22851C(1), type IIA, shall be pumped through the pumping system to test the ability of the system to produce flow rates conforming to the requirements of 3.8.3.4.

4.4.7 Drum unloading. The drum suction hose and pipe (see 3.8.5), shall be installed in accordance with the manufacturer's recommendations. The pumping system shall demonstrate its ability to pump oil from drums located at ground level. A minimum of 100 gallons of aircraft lubricating oil, conforming to MIL-L-22851C(1), type IIA, shall be pumped from the drums into the oil dispensing tank to determine conformance of 3.8.3.4 and 3.8.5.

4.4.8 Heating system. The heating system, when furnished, shall be operated for a sufficient length of time to insure proper functioning of the system as specified in 3.8.6.

4.4.9 Failure. Failure of the first production vehicle to meet requirements of the contract shall be cause for the Government to refuse acceptance of all vehicles under the contract until corrective action has been taken.

4.4.10 Production sample. Upon acceptance of the first production vehicle, it shall remain at the manufacturing facility as a production sample, and be the last truck shipped on the contract. The first production vehicle shall be reconditioned, including replacement of abnormally worn parts and paint touch-up or repainting, prior to delivery to enable it to be accepted as a contract item. The contractor shall maintain the first production vehicle in a serviceable condition for the duration of the contract.

4.5 Production unit tests. The contractor's testing system shall, as a minimum, assure that the fully equipped truck is capable of meeting the performance requirements specified herein. Tank shall be subjected to the pressure tests specified in 4.4.2.

4.6 Packaging inspection. The vehicle shall be inspected to verify conformance to the requirements of section 5.

5. PACKAGING

5.1 Vehicle processing. The equipment shall be preserved and packaged in accordance with the contractor's standard practice. When specified (see 6.2), equipment shall be preserved and packaged in accordance with the mobile requirements of MIL-STD-281 with the level of preservation and packaging as specified (see 6.2).

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6. NOTES

6.1 Intended use. The aircraft lubricating oil dispensing truck covered by this specification is intended for use at established Naval Stations for transporting and dispensing aviation lubricating oil to aircraft, both within and without the continental limits of the United States. They will be operated over improved and unimproved roads having grades usually found on main and access roads in and about established naval bases.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. When powerplant heaters are required (see 3.7.1.3).
c. When heating system is required (see 3.8.6).
d. When rustproofing is required (3.12).
e. When preservation and packaging in accordance with MIL-STD-281 is required, and the level of preservation required (see 5.1).

6.3 Performance prediction. Completed copies of SAE Truck Ability Prediction Procedure form and computation for low speed requirement for each vehicle model furnished under contract should be submitted as specified in the contract. The SAE Work Sheet Item 1 should include vehicle model number, wheelbase, engine model number, transmission model number, and vehicle type. Unless other conditions are cited in the contract, computations should be made for normal atmospheric pressure, normal ambient air temperature, and still, dry air. The factors to be used in predicting truck ability (see 3.6.1.1) are established as following for the corresponding SAE Truck Ability Prediction Procedure Tables:

Table 1 - Tire Factor. This factor must relate to the size of tires furnished by the contractor in accordance with this specification.
Table 2 - Altitude Factor. 1.00.
Table 3 - Rolling Factor. 1.613.
Table 4 - Area Factor. 0.173.
Table 5 - Velocity Factor. 250.0.
Table 6 - Altitude Factor. 1.00.
Table 7 - Chassis Friction Horsepower. Use applicable power unit gvw (to nearest, higher, 1,000 lb) and the engine rpm (to nearest 100 revolutions) which is required for 50 mph geared speed. For gvw and engine speed beyond the range of this table, factors shall be extrapolated.
Table 8 - Grade Factor. 0.75.
Table 8A - Correction Factor. Not required.
Table 9 - Road Factor. 0.0.

6.4 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any

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inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodian:

Navy - YD

User activity:

Navy - MC

Preparing activity:

Navy - YD

(Project 2320-NC38)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

2320-NC38

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-T-21900F(YD)	TRUCK, TANK; AIRCRAFT LUBRICATING OIL LUBRICATING OIL DISPENSING, 500 GALLON, 4 BY 2

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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